T.S. of Leaflet

The leaflet of *Cycas* in transverse section shows the presence of upper and lower epidermis. The epidermal cells are thick walled and are covered with thick cuticle. The lower epidermis is not continuous and is interrupted by **sunken stomata**. The hypodermis consists of sclerenchyma cells to prevent transpiration. The mesophyll is differentiated into palisade and spongy parenchyma. The cells of this layer are involved in photosynthesis. The spongy parenchyma present in close proximity to the lower epidermis bear large intercellular spaces which help in gaseous exchange.

Layers of colourless, elongated cells which run parallel to the leaf surface from the midrib to the margin of the leaflet are seen. These constitute the **Transfusion tissue** that helps in the lateral conduction of water. The vascular bundle has xylem facing upper epidermis and phloem facing lower epidermis. The protoxylem occupies the centre, hence the bundle is mesarch. The vascular bundle has a sclerenchymatous bundle sheath (Figure 2.28).

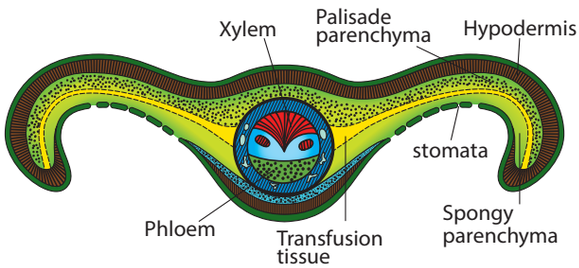


Figure 2.28: T.S. of leaflet

Reproduction

Cycas reproduces by both vegetative and sexual methods.

Vegetative reproduction

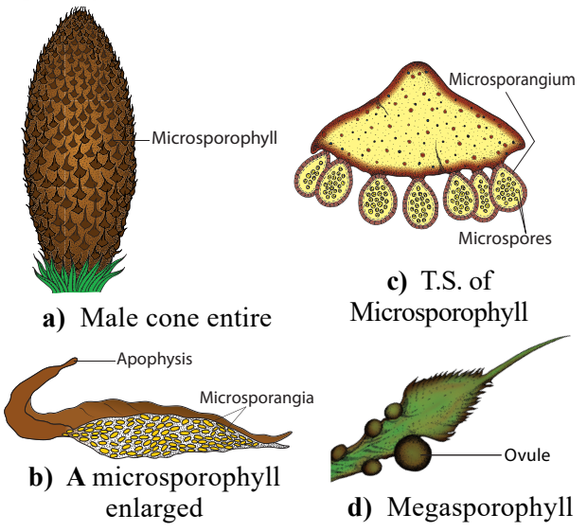
It takes place by adventitious buds or bulbils. They develop in the basal part of the stem. The bulbils on germination produce new plants.

Sexual reproduction

Cycas is dioecious i.e., male and female cones are produced in separate plants. It is heterosporous and produces two types of spores (Figure 2.29).

Male cone

The male cone or staminate cone are borne singly on the terminal part of the stem. The growth of the stem is continued by the formation of axillary buds at the base of the cone. The male cone is displaced to one side showing sympodial growth in the stem. Male cones are stalked, compact, oval or conical and woody in structure. It consists of several microphylls which are arranged spirally around a central cone axis.



a) Male cone entire

b) **A** microsporophyll enlarged

c) T.S. of Microsporophyll

d) Megasporophyll

Figure 2.29: Reproduction in *Cycas*

Microsporophylls

Microsporophylls are flat, leaf-like and woody structures with narrow base and expanded upper portion. The upper expanded portion becomes pointed and is called apophysis. The narrow base is attached to the cone axis. Each microsporophyll contains thousands of microsporangia in groups called sori on abaxial (lower) surface. Development of sporangium is of eusporangiate type. The spore mother cell undergoes meiosis to produce halpoid microspores. Each microsporangium bears large number of microspores or pollen grains. Each microspore (Pollen grain) is a rounded, unicellular and uninucleate structure surrounded by outer thick exine and an inner thin intine. The microspore represents the male gametophyte.

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